Problems, Programs, and Ideal Complexity

1 Problems

1.1 Definition of Problem $\mathcal{Q}$

A problem is also referred to as a "Question"

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{Q} := f[X_n] \rightarrow A_o \in \Omega \quad \forall X_n \in \mathbb{D}_f$$

1.2 True False Question

Also referred to as a "Decision Question", "Decision Problem" xor "True False Problem"

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{Q} := f[X_n] \rightarrow y_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_f$$

1.3 Numerical Problem

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{Q} := f[X_n] \rightarrow y_o \in \mathbb{C} \quad \forall X_n \in \mathbb{D}_f$$

1.4 System Question

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{Q} := f[X_n] \rightarrow A_o \in \Omega \quad \forall X_n \in \mathbb{D}_f :$$
$$|A_o| == M$$

2 Instructions

2.1 Definition of a Canonical Instruction Set

Also denoted as an Instruction Set Architecture (ISA)

2.2 Definition of Instruction

An instruction is either a canonical comparison, canonical operation, or canonical measurement of Question $\mathcal{Q}$ given ISA $\mathcal{A}$ followed by an assignment

2.3 Definition of Instruction Tree

Define $\mathcal{T}$; a tree of of instructions $\tilde{s}_i$

$$\mathcal{T} := \{\tilde{s}_1, \tilde{s}_i, ...\}$$

2.4 Definiton of a Closed Path in an Instruction Tree

2.5 Definition of the Tensor of Closed Paths in an Instruction Tree

2.6 Definition of Valid Instruction Path of an Instruction Tree

Define Instruction Path $\mathcal{I}$ of Instruction Tree $\mathcal{T}$; a closed path of ordered instructions s_i

Let Instruction Path $\mathcal{I}$ of Instruction Tree $\mathcal{T}$

$$\mathcal{I}_{\mathcal{T}} := \{s_1, s_2, ..., s_N\} :$$
$$(s_1 == \tilde{s}_1) \wedge ((s_i \ -> s_{i+1}) \in \mathcal{T} \quad \forall i < N) \wedge (s_N == return\ y)$$

3 Memory

3.1 Definition of Memory Set

Define Memory $\mathcal{M}$; an ordered set of objects m_i

$$\mathcal{M} := \{m_1, m_2, ..., m_M\}$$

4 Programs

4.1 Definition of a Program

Define Program P

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{P} := P[X_n] \to y_o \in \Omega \quad \forall X_n \in \mathbb{D}_P =$$
$$\{s_1, s_2, ..., s_N \,|y_o, b_2, b_3, ..., b_M\} \to y_o \in \Omega \quad \forall X_n \in \mathbb{D}_P$$

4.2 Void Programs

Define a void program; a program with inputs x_i and no output

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{P} := P[X_n] \quad \forall X_n \in \mathbb{D}_P =$$
$$\{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M\} \quad \forall X_n \in \mathbb{D}_P$$

4.3 Boolean Programs

Define a Boolean Program
Given

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{P} := P[X_n] \to y_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_P \quad =$$
$$\{s_1, s_2, ..., s_N \mid b_2, b_3, ..., b_M, y_o\} \to y_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_P$$

4.4 Numerical Programs

Define a numerical program; a program with inputs x_i, input set C, and real, rational output y_o

$$X_n = \{x_1, ..., x_n\}$$
$$\mathcal{P} := P[X_n] \to y_o \in \mathbb{Q} \wedge y_o \geqslant 0 \quad \forall X_n \in \mathbb{D}_P =$$
$$\{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, y_o\} \to y_o \in \mathbb{Q} \wedge y_o \geqslant 0 \quad \forall X_n \in \mathbb{D}_P =$$

4.5 System Programs

Define a system program; a program with inputs x_i, input set C, and real, output set Y_o

$$X_n = \{x_1, ..., x_n\}$$
$$\mathcal{P} := P[X_n] \to Y_o \in \Omega = \{y_1, y_2, ..., y_K\} \in \Omega \quad \forall X_n \in \mathbb{D}_P \quad =$$
$$\{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, Y_o\} \to Y_o \in \Omega = \{y_1, y_2, ..., y_K\} \in \Omega \quad \forall X_n \in \mathbb{D}_P$$

4.6 Domain of a Program

The space of inputs/outputs

5 Program Complexity

5.1 Definition of Computational Complexity of Serial Program $\mathcal{P}$

Define computational complexity of Program $\mathcal{P}$ with dimension C

$$\mathbf{O}[n] := \ < O_T[n], O_S[n], O_3[n], ..., O_C[n] >$$

5.2 Definition of Ideal Computational Complexity of Serial Program $\mathcal{P}$

Define ideal computational complexity of Program $\mathcal{P}$ with dimension 2

$$\mathbf{O}[n] := \ < O_T[n], O_S[n] >$$

5.3 Assumption 1

The remainder of this document assumes complexity means ideal computational complexity with dimension 2.

5.4 Total Complexity

Define Total Complexity of Boolean Program $\mathcal{P}$

$$O[n] := |P[X_n]| = |\{\mathcal{L}, \mathcal{M}, y_o\}|$$

$$= |\mathcal{L}| + |\mathcal{M}| + |y_o| = N + M$$

5.5 Time Complexity

Define Time Complexity $O_T[n]$ of Boolean Program $\mathcal{P}$

$$\mathcal{P} = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_T[n] := |\mathcal{L}| = N$$

5.6 Space Complexity

Define Space Complexity $O_S[n]$ of Boolean Program $\mathcal{P}$

$$s^+ = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$O_S[n] := |\mathcal{M}| + |y_o| = M$$

5.7 Total Complexity as a Function of Time and Space Complexity

$$\begin{aligned} O[n] := |s^+[X_n]| &= |\{\mathcal{L}, \mathcal{M}, y_o\}| \\ &= |\mathcal{L}| + |\mathcal{M}| + |y_o| \\ &= O_T[n] + O_S[n] \end{aligned}$$

5.8 $O_S[n] > 0^*$

Assuming Program is not void

5.8.1 Proof

Assume $O_S[n] = 0$

$$O_S[n] := |\mathcal{M}| + |y_o|$$
$$O_S[n] = 0 \Rightarrow \mathcal{M} = y_o = \varnothing$$
$$y_o = \varnothing; \;\; y_o \in \{\mathbb{T}, \mathbb{F}\} \text{ by definition of } s^+$$

$\therefore O_S[n] = 0$ contradicts the definition of solution s^+ of a decision problem

$$O_S[n] \geqslant 0 \text{ by definition of magnitude}$$
$$\therefore O_S[n] > 0$$

5.9 $O_T[n] > 0^*$

Assuming Program is not void

5.9.1 Proof

Assume $O_T[n] = 0$

$$O_T[n] := |\mathcal{L}|$$

$$O_T[n] = 0 \Rightarrow y_o \notin \{\mathbb{T}, \mathbb{F}\}$$

$$y_o \notin \{\mathbb{T}, \mathbb{F}\}; \;\; y_o \in \{\mathbb{T}, \mathbb{F}\} \text{ by definition of } s^+$$

$$\therefore O_T[n] = 0 \text{ contradicts the definition of solution } s^+ \text{ of a decision problem}$$

$$O_T[n] \geqslant 0 \text{ by definition of magnitude}$$

$$\therefore O_T[n] > 0$$

5.10 $O[n] > 0^*$

Assuming Program is not void

5.10.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_T[n] > 0; \;\; O_S[n] > 0$$

$$\therefore O[n] > 0$$

5.11 $O[n] > O_T[n]^*$

Assuming Program is not void

5.11.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_S[n] > 0$$

$$\therefore O[n] > O_T[n]$$

5.12 $O[n] > O_S[n]^*$

Assuming Program is not void

5.12.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_T[n] > 0$$

$$\therefore O[n] > O_S[n]$$

5.13 $O[n+1] \geqslant O[n]$

5.13.1 Proof

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_{n+1}, C\}$$

$$O[n] = |\, s^+[X_i]|$$
$$O[n+1] = \hat{O}[n] = |\, s^+[\hat{X}_i]|$$

For general solutions s^+

$$s^+[\hat{X}_i] \supseteq s^+[X_i]$$
$$\Rightarrow |s^+[\hat{X}_i]| \geqslant |s^+[X_i]|$$
$$\therefore \hat{O}[n] = O[n+1] \geqslant O[n]$$

6 Worst Case Total Complexity $O_{wc}[n]$

Let Program $\mathcal{P}$ with input space $X_n = \{x_1, x_2, ..., x_n\}$ and number of inputs n $= \tilde{n}$

$$O_{wc}[\tilde{n}] := O_T[\tilde{n}] : O_T[\tilde{n}] \geqslant O_{T_i}[\tilde{n}] \quad \forall X_{\tilde{n}} \in \mathbb{D}_{\mathcal{P}}$$

6.1 Proof of Existence

Let Program $\mathcal{P}$ with input space $\mathbb{X}_n$ having magnitude > 0
Prove the existence of $O_{wc}[n]$

7 Total Runtime Complexity

8 Control Flow Statements

If, then, while, enum

8.1 Early return statements

9 Proving Wost Case Complexity as a function of n

Given ISA $\mathcal{A}$, when is it provable that O_{wc} is a function of n?

10 Inductive Complexity Expressions

$$O[n] = O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1]$$

$$f_{n+1}[n] := O[n+1] - O[n]$$

10.1 Inductive Space and Time Formulas

$$f^T_{n+1}[n] := O_T[n+1] - O_T[n]$$
$$O_T[n+1] = O_T[n] + f^T_{n+1}[n]$$
$$f^S_{n+1}[n] := O_S[n+1] - O_S[n]$$
$$O_S[n+1] = O_S[n] + f^S_{n+1}[n]$$

10.2 Inductive Function Expressions

Relate $f_{n+1}[n]$ to equivalence functions

$$O[n] = O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1] = O[n] + f_{n+1}[n]$$
$$O_T[n] = O[n] - O_S[n]$$
$$O_S[n] = O[n] - O_T[n]$$

$$f_{n+1}[n] = O[n+1] - O[n]$$
$$f_{n+1}[n] = O_T[n+1] + O_S[n+1] - O[n]$$
$$f_{n+1}[n] = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n]$$
$$f_{n+1}[n] = O[n+1] - O_T[n] - O_S[n]$$
$$f_{n+1}[n] = f^T_{n+1}[n] + f^S_{n+1}[n]$$

10.3 Zero Order Space Inductive Function

$$Let\ \ O_S[n] \sim n^0$$
$$f_{n+1}[n] = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n] = O_T[n+1] - O_T[n]$$

11 The Set of all Programs

$$\Omega_{\mathcal{P}} := \{\mathcal{P}_i, ...\} \quad \forall i$$

12 Polynomial Complexity

12.1 Definition of a Program with Polynomial Complexity

Boolean Program $\mathcal{P}$ with worst case total complexity $O_{wc}[n]$ is said to be *polynomial bounded* if and only if

$$\exists K, \lambda_0, \lambda_1 ... \lambda_K \quad :$$
$$O_{wc}[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} + ... + \lambda_1 n + \lambda_0 \quad \forall n$$

12.2 Polynomial Order of Complexity

Boolean Program $\mathcal{P}$ with worst case total complexity $O_{wc}[n]$ is said to be on the order of its maximum polynomial coefficient

$$\exists K, \lambda_0, \lambda_1 ... \lambda_K :$$
$$O_{wc}[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + \lambda_0 \quad \forall n$$
$$\Rightarrow O_{wc}[n] \sim n^K$$

12.3 The set of Polynomial Programs

Define $\mathbb{P}$, the set of Programs having polynomial worst case complexity

$$\mathbb{P} := \{\mathcal{P}_1, \mathcal{P}_2, ...\} :$$

$$\mathcal{P}_i \textit{ is polynomial bounded} \quad \forall P_i \in \mathbb{P}$$

12.4 Property of Polynomial Complexity 1

$$\mathcal{P} \textit{ is polynomial bounded}$$

$$\Leftrightarrow$$

$$lim_{n \to \infty} \frac{O_{wc}[n+1]}{O_{wc}[n]} = 1$$

12.4.1 Proof

$$O_{wc}[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + \lambda_0$$

$$O_{wc}[n+1] = (\lambda_K (n+1))^K + (\lambda_{K-1}(n+1))^{K-1} ... + \lambda_1 (n+1) + \lambda_0$$
$$= (\lambda_K n)^K + (\tilde{\lambda}_{K-1} n)^{K-1} ... + \tilde{\lambda_1} n + \tilde{\lambda}_0$$

$$lim_{n\to\infty} \frac{O_{wc}[n+1]}{O_{wc}[n]}$$

$$= lim_{n\to\infty} \frac{(\lambda_K n)^K + (\tilde{\lambda}_{K-1} n)^{K-1} ... + \tilde{\lambda_1} n + \tilde{C}}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C}$$

$$= lim_{n\to\infty} \frac{(\lambda_K n)^K}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C} + \frac{(\tilde{\lambda}_{K-1} n)^{K-1}}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C} + ... +$$
$$\frac{\tilde{\lambda}_1 n}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C} + \frac{\tilde{C}}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C}$$

$$= 1 = lim_{n\to\infty} \frac{O_{wc}[n+1]}{O_{wc}[n]}$$

12.5 Property of Polynomial Complexity 2

$$\exists K, \hat{\lambda}_0, \hat{\lambda}_1, ..., \hat{\lambda}_K :$$
$$O[n+1] - O[n] = f_{n+1}[n] = (\hat{\lambda}_{K-1} n)^{K-1} ... + \hat{\lambda}_1 n + \hat{\lambda}_0 \quad \forall n$$

12.5.1 Proof

$$O_{wc}[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + \lambda_0$$

$$O_{wc}[n+1] = (\lambda_K (n+1))^K + (\lambda_{K-1}(n+1))^{K-1} ... + \lambda_1 (n+1) + \lambda_0$$
$$= (\lambda_K n)^K + (\tilde{\lambda}_{K-1} n)^{K-1} ... + \tilde{\lambda_1} n + \tilde{\lambda}_0$$

$$O_{wc}[n+1] - O_{wc}[n] =$$
$$(\lambda_K n)^K - (\lambda_K n)^K + (\tilde{\lambda}_{K-1}^{K-1} - \lambda_{K-1}^{K-1}) n^{K-1} ... + (\tilde{\lambda_1} - \lambda_1) n + (\lambda_1 - \lambda_0)$$

$$O_{wc}[n+1] - O_{wc}[n] = (\hat{\lambda}_{K-1} n)^{K-1} ... + \hat{\lambda}_1 n + \hat{\lambda_0}$$

12.6 Property of Polynomial Complexity 3

$$limit_{n\to\infty} \frac{f_{n+1}[n]}{O[n]} = 0$$

12.6.1 Proof

$$limit_{n\to\infty}\frac{O[n+1]}{O[n]} = 1$$

$$limit_{n\to\infty}\frac{O[n]+f_{n+1}[n]}{O[n]} = 1$$

$$limit_{n\to\infty}\frac{O[n]}{O[n]} + \frac{f_{n+1}[n]]}{O[n]} = 1$$

$$limit_{n\to\infty}1 + \frac{f_{n+1}[n]]}{O[n]} = 1$$

$$limit_{n\to\infty}\frac{f_{n+1}[n]]}{O[n]} = 0$$

12.7 Total Polynomial Complexity Implies Time bounded Polynomial Complexity

$$D \in \mathbb{P} \Longrightarrow O_T[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$

12.7.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \ \ \forall n$$

$$O[n] := O_T[n] + O_S[n]; \ \ O_S[n] > 0$$

$$\therefore O_T[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \ \ \forall n$$

12.8 Total Polynomial Complexity Implies Space bounded Polynomial Complexity

$$D \in \mathbb{P} \Longrightarrow O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$

12.8.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \ \ \forall n$$

$$O[n] := O_T[n] + O_S[n]; \ \ O_T[n] > 0$$

$$\therefore O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \ \ \forall n$$

12.9 Polynomial Complexity in Space and Time Implies Polynomial Total Complexity

$$(O_S[n] == (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + \lambda_0)$$

$$\wedge$$

$$(O_T[n] == (\hat{\lambda}_M n)^M + (\hat{\lambda}_{M-1} n)^{M-1} ... + \hat{\lambda}_1 n + \hat{\lambda}_0)$$

$$\Rightarrow D \in \mathbb{P}$$

12.9.1 Proof

$$O_S[n] = \lambda_K n^K + \lambda_{K-1} n^{K-1} + ... + \lambda_1 n + \lambda_0$$

$$O_T[n] = \hat{\lambda}_M n^M + \hat{\lambda}_{M-1} n^{M-1} + ... + \hat{\lambda}_1 n + \hat{\lambda}_0$$

$$O[n] = O_S[n] + O_T[n]$$

$$^*O[n] = (\hat{\lambda}_0 + \lambda_0) + n(\lambda_1 + \hat{\lambda_1}) + ... + n^K(\lambda_K + \hat{\lambda_K}) + \hat{\lambda}_{K+1} n^{K+1} + ... + \hat{\lambda}_M n^M$$

$$\therefore O[n] \text{ has polynomial total complexity by definition}$$

* Assume K $<$ M, similar proof for K=M, K>M

13 Non-Polynomial Complexity

13.1 Definition

Boolean Program $\mathcal{P}$ is said to have *non polynomial complexity* if and only if

$$\nexists K, \lambda_0, \lambda_1, ..., \lambda_K \ :$$
$$O_{wc}[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} + ... + \lambda_1 n + \lambda_0 \quad \forall n$$

13.2 Non-Polynomial Problems

Define $\mathbb{N}$, the set of Programs that do not have Polynomial Total Complexity

$$\mathbb{N} := \{\mathcal{P}_1, \mathcal{P}_2, ...\} :$$
$$\nexists K, \lambda_0, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + \lambda_0 \quad \forall n, \ \ \forall \mathcal{P}_i \in \mathbb{N}$$

13.3 Universe of Programs Theorem

Prove

$$\mathbb{P} \cup \mathbb{N} = \Omega_{\mathcal{P}}$$

13.3.1 corrolary 1

$$\mathbb{N} = \Omega_{\mathcal{P}} \backslash \mathbb{P}$$

13.3.2 corrolary 2

$$\mathbb{P} = \Omega_{\mathcal{P}} \backslash \mathbb{N}$$

13.4 $\mathbb{P}$ and $\mathcal{N}$ are disjoint

$$\mathbb{P} \cap \mathcal{N} = \varnothing$$

13.4.1 Proof

Let D $\in \mathcal{N}$

$$\nexists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Assume D $\in \mathbb{P}$

$$\exists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Contradicts the definition of $\mathcal{N}$

$$\therefore D \in \mathcal{N} \Rightarrow D \notin \mathbb{P}$$

Let D $\in \mathbb{P}$

$$\exists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Assume D $\in \mathcal{N}$

$$\nexists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Contradicts the definition of $\mathbb{P}$

$$\therefore D \in \mathbb{P} \Rightarrow D \notin \mathcal{N}$$

$$D \in \mathcal{N} \Rightarrow D \notin \mathbb{P}; D \in \mathbb{P} \Rightarrow D \notin \mathcal{N}$$
$$\therefore \mathbb{P} \cap \mathcal{N} = \varnothing$$

14 Divergent Complexity

See Appendix for additional proofs on Convergent and Divergent Functions

14.1 Divergent Programs

See Appendix for additional proofs on Convergent and Divergent Functions

14.2 Divergent Program $\Rightarrow$ Not-Polynomial Program

Enter proof

14.3 Not-Polynomial Program $\Rightarrow$ Divergent Programs

Disprove xor prove

15 Solution to a Problem

15.1 Definition of Functionally Contains

Let

$$f_1[X] \to a_1 \in \Omega \quad \forall X \in \mathbb{D}_{f_1}$$
$$f_2[\hat{X}] \to a_2 \in \Omega \quad \forall \hat{X} \in \mathbb{D}_{f_2}$$

$$(f_1[] \to a_1 \overset{f}{\subseteq} f_2[] \to a_2) \to y \in \{\mathbb{T}, \mathbb{F}\} :=$$
$$(\mathbb{D}_2 \subseteq \mathbb{D}_1) \wedge (a_1 == a_2 \quad \forall \hat{X} \in \mathbb{D}_2)$$

15.2 Definition of Functionally Equivalent

Let

$$f_1[X] \to a_1 \in \Omega \quad \forall X \in \mathbb{D}_{f_1}$$
$$f_2[X] \to a_2 \in \Omega \quad \forall X \in \mathbb{D}_{f_2}$$

$$(f_1 \overset{f}{==} f_2) \to y \in \{\mathbb{T}, \mathbb{F}\} :=$$
$$(a_1 == a_2 \quad \forall X \in \mathbb{D}_1) \wedge (\mathbb{D}_1 == \mathbb{D}_2)$$

15.3 Definition of a Solution of Boolean Problem

Let Boolean Problem Q

$$\mathcal{Q} := f[X_n] \to a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_Q$$

$$S_Q^+[X_n] := P[X_n] \to y = \{s_1, s_2, ..., s_N \mid y, b_2, ..., b_M\} \to y :$$
$$(P[X_n] \to y) \wedge (y == a_o) \quad \forall X_n \in \mathbb{D}_Q$$

15.4 Complexity of a Solution

Let Boolean Problem $\mathcal{Q}$ having solution $S^+ \in \mathbb{S}^+$

$$\mathcal{Q} := f[X_n] \to a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_Q$$

$$S_Q^+[X_n] := P[X_n] \to y = \{s_1, s_2, ..., s_N \mid y, m_2, ..., m_M\} \to y :$$
$$(P[X_n] \to y) \wedge (y == a_o) \quad \forall X_n \in \mathbb{D}_Q$$

$$S_Q^+[X_n] \to y \quad \forall X_n \in \mathbb{D}_Q$$
$$= P[X_n] \to y = \{s_1, s_2, ..., s_N \mid y, m_2, ..., m_M\} \to y \quad \forall X_n \in \mathbb{D}_Q$$
$$= \{\mathcal{I}, \mathcal{M}\} \to y \quad \forall X_n \in \mathbb{D}_Q$$

$$Complexity[S_Q^+] := |\mathcal{I}| + |\mathcal{M}| = N + M$$

15.4.1 Proof of existence of Complexity of Solution

By definition S^+ is a Program, $\mathcal{I}$ and $\mathcal{M}$ are sets, therefore $|\mathcal{I}|$ *and* $|\mathcal{M}|$ are defined

15.5 Worst Case Complexity of Solution s_{wc}^+

15.6 Proving the existence of s_{wc}^+

15.7 Solution Space of a Boolean Problem

Let Boolean Problem Q

$$\mathcal{Q} := f[X_n] \to y_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_Q$$

$$\mathbb{S}_Q^+ := \{S_i^+, ...\} :$$
$$S_i^+[X_n] := P[X_n] = \{s_1, s_2, ..., s_N \mid y, b_2, ..., b_M\} :$$
$$(P_Q[X_n] \to y) \wedge (y == a) \quad \forall X_n \in \mathbb{D}_Q$$
$$\forall S_i^+ \in \mathbb{S}^+$$

15.8 Candidate Solutions

16 Subfunctions

17 Subfunction Decomposition of Program

18 Subsolutions

18.1 Subfunction condition of solutions

Prove (xor state as a formal condition)

$$P[X_{n+1}] \supset P[X_n]$$

19 Inductive Complexity Expressions

$$O[n] = O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1]$$

$$f_{n+1}[n] := O[n+1] - O[n]$$

19.1 Inductive Space and Time Formulas

$$f^T_{n+1}[n] := O_T[n+1] - O_T[n]$$
$$O_T[n+1] = O_T[n] + f^T_{n+1}[n]$$
$$f^S_{n+1}[n] := O_S[n+1] - O_S[n]$$
$$O_S[n+1] = O_S[n] + f^S_{n+1}[n]$$

19.2 Inductive Function Expressions

Relate $f_{n+1}[n]$ to equivalence functions

$$O[n] = O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1] = O[n] + f_{n+1}[n]$$
$$O_T[n] = O[n] - O_S[n]$$
$$O_S[n] = O[n] - O_T[n]$$

$$f_{n+1}[n] = O[n+1] - O[n]$$

$$f_{n+1}[n] = O_T[n+1] + O_S[n+1] - O[n]$$
$$f_{n+1}[n] = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n]$$
$$f_{n+1}[n] = O[n+1] - O_T[n] - O_S[n]$$
$$f_{n+1}[n] = f^T_{n+1}[n] + f^S_{n+1}[n]$$

19.3 Zero Order Space Inductive Function

$$Let\ \ O_S[n] \sim n^0$$
$$f_{n+1}[n] = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n] = O_T[n+1] - O_T[n]$$

20 Discrete Derivative; Z Transform

20.1 Discrete Derivative

Define derivative for discrete function f[n]

$$\Delta_n^1 f[n] := f[n+1] - f[n]$$

We will use the above definition for the remainder of this document

20.2 Zero Order Derivative

$$\Delta_n^0 f[n] = f[n]$$

20.3 Kth Discrete Derivative

Define the Kth derivative of discrete function $f[n]$

$$\Delta_n^K f[n] := \Delta_n^{K-1} f[n+1] - \Delta_n^{K-1} f[n]$$

20.4 Kth Discrete Derivative as an Alternating Sum

$$\Delta_n^K f[n] := \Delta_n^{K-1} f[n+1] - \Delta_n^{K-1} f[n]$$

$$= (\Delta_n^{K-2} f[n+2] - \Delta_n^{K-2} f[n+1]) - (\Delta_n^{K-2} f[n+1] - \Delta_n^{K-2} f[n])$$

$$= (\Delta_n^{K-2} f[n+2] - 2\Delta_n^{K-2} f[n+1] - \Delta_n^{K-2} f[n])$$

$$= \textstyle\sum_{i=0}^{K} (-1)^j \ ({}_K C_j) \ \Delta_n^0 f[n+j]$$

$$= \textstyle\sum_{i=0}^{K} (-1)^j \ ({}_K C_j) \ f[n+j]$$

20.5 Z Transform

Define the Z Transform for discrete function f[n]

$$\mathcal{Z}(f[n]) := \textstyle\sum_{n=0}^{\infty} f[n] z^{-n}$$

20.6 Z Transform of 0 Order Derivative

$$\Delta_n^0 f[n] := f[n]$$

$$\mathcal{Z}(\Delta_n^0 f[n]) = \mathcal{Z}(f[n])$$

20.7 Z Transform of 1st Derivative

$$\Delta_n^1 f[n] := f[n+1] - f[n]$$

$$\mathcal{Z}(\Delta_n^1 f[n]) = \mathcal{Z}(f[n+1] - f[n])$$

$$= \sum_{n=0}^{\infty}(f[n+1] - f[n])z^{-n}$$

$$= \sum_{n=0}^{\infty}(f[n+1]z^{-n} - f[n]z^{-n})$$

$$= \sum_{n=0}^{\infty} f[n+1]z^{-n} - \sum_{n=0}^{\infty} f[n]z^{-n}$$

$$= \sum_{m=0}^{\infty} f[m+1]z^{-m} - \sum_{n=0}^{\infty} f[n]z^{-n}$$

Let

$$\hat{m} = m + 1; \;\; m = \hat{m} - 1$$

$$= \sum_{m=0}^{\infty} f[\hat{m}]z^{-(\hat{m}-1)} - \mathcal{Z}(f[n])$$

$$= z^1 \sum_{\hat{m}=1}^{\infty} f[\hat{m}]z^{-\hat{m}} - \mathcal{Z}(f[n])$$

$$= z^1 \sum_{\hat{m}=1}^{\infty} f[\hat{m}]z^{-\hat{m}} + f[0] - f[0] - \mathcal{Z}(f[n])$$

$$= z^1 \sum_{\hat{m}=0}^{\infty} f[\hat{m}]z^{-\hat{m}} - f[0] - \mathcal{Z}(f[n])$$

$$= z^1 \mathcal{Z}(f[n]) - f[0] - \mathcal{Z}(f[n])$$

$$\mathcal{Z}(\Delta_n^1 f[n]) = \mathcal{Z}(f[n])(z^1 - 1) - f[0]$$

20.8 Z Transform of Kth Derivative

$$\mathcal{Z}(f[n]) := \sum_{n=0}^{\infty} f[n]z^{-n}$$

$$\mathcal{Z}(\Delta_n^K f[n]) = \sum_{n=0}^{\infty} \Delta_n^K f[n]z^{-n}$$

$$= \sum_{n=0}^{\infty} \sum_{i=0}^{K} (-1)^j \; ({}_K C_j) \; f[n+j]z^{-n}$$

$$= \sum_{n=0}^{\infty}(f[n+K] - ({}_K C_1)f[n+K-1] + ({}_K C_2)f[n+K-2] - ... \pm f[n])z^{-n}$$

$$= \sum_{n=0}^{\infty} f[n+K]z^{-n} - ({}_K C_1)f[n+K-1]z^{-n} + ({}_K C_2)f[n+K-2]z^{-n} - \\ ... \pm f[n]z^{-n}$$

$$= z^K \mathcal{Z}(f[n]) + \sum_{i=0}^{K-1} f[i] - ({}_K C_1)z^{K-1}\mathcal{Z}(f[n]) - \sum_{j=0}^{K-2} f[j] + \\ ({}_K C_2)z^{K-2}\mathcal{Z}(f[n]) + \sum_{k=0}^{K-3} f[k] - ... \pm \mathcal{Z}(f[n])$$

When K is odd

$$\mathcal{Z}(\Delta_n^K f[n]) = (z-1)^K \mathcal{Z}(f[n]) + \sum_{i=0}^{\frac{n+1}{2}} f[2i] \quad K > 0$$

When K is even

$$\mathcal{Z}(\Delta_n^K f[n]) = (z-1)^K \mathcal{Z}(f[n]) + \sum_{j=0}^{\frac{n}{2}} f[2j+1] \quad K > 0$$

Computation Basis

21 Definition of Basis

22 Definition of Necessary Instructions

$$s_i \in S_j^+ \quad \forall S_j^+ \in \mathbb{S}_Q$$

23 Definition of Convertible Instructions

$$s_i \in S_j^+ \quad "for\ some"\ S_j^+ \in \mathbb{S}_Q$$

24 Theorem of Polynomial Subfunctions

25 Theorem of Divergent Subfunctions

26 Two integer Sum to N

26.1 State formal definition of Sum to N : $x_i + x_j == N$

$$X_n = \{x_1, ..., x_n\}$$

$$D := f[X_i, N] = a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$

$$s^+[X_n] = P[X_n] :$$
$$(P[X_i] = y_o == a_o \quad \forall X_i) \quad \cap \quad (P[X_{n+1}] \supseteq P[X_n] \quad \forall X_{n+1})$$

$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$D = f[X_i] = \quad \exists x_j, x_k \in X_n \quad j \neq k :$$
$$x_j + x_k == N$$

26.2 Express a formal solution : $O_S[n] \sim n^0$

$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$s_1 = y_o \leftarrow \mathbb{F};$$
$$\forall i < n \ , \ n \geqslant j > i$$
$$s_2, s_3, s_8, s_9, ..., s_{3ij-4}, s_{3ij-3}..., s_{3n(n-1)-4}, s_{3n(n-1)-3} = b_1 \leftarrow x_i + x_j$$
$$s_4, s_5, s_{10}, s_{11}, ..., s_{3ij-2}, s_{3ij-1}..., s_{3n(n-1)-2}, s_{3n(n-1)-1} = b_1 \leftarrow b_1 == N$$
$$s_6, s_7, s_{12}, s_{13}..., s_{3ij}, s_{3ij+1}..., s_{3n(n-1)}, s_{3n(n-1)+1} = y_o \leftarrow y_o \vee b_1$$
$$s^+ = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\}$$

26.3 Prove s^+ satisfies the subfunction condition of solutions: $P[X_{n+1}] \supseteq P[X_n] \quad \forall X_{n+1}$

$$X_n = \{x_1, x_2, ..., x_n\}; \quad X_{n+1} = \{x_1, x_2, ..., x_n, x_{n+1}\}$$
$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$s^+_{n+1} = s^+ \cup \hat{s}^+$$
$$s_1 = y_o \leftarrow \mathbb{F};$$
$$\forall i < n \ , \ n \geqslant j > i$$

$$s_2, s_3, s_8, s_9, ..., s_{3ij-4}, s_{3ij-3}..., s_{3n(n-1)-4}, s_{3n(n-1)-3} = b_1 \leftarrow x_i + x_j$$
$$s_4, s_5, s_{10}, s_{11}, ..., s_{3ij-2}, s_{3ij-1}..., s_{3n(n-1)-2}, s_{3n(n-1)-1} = b_1 \leftarrow b_1 == N$$
$$s_6, s_7, s_{12}, s_{13}..., s_{3ij}, s_{3ij+1}..., s_{3n(n-1)}, s_{3n(n-1)+1} = y_o \leftarrow y_o \vee b_1$$

$$\forall k < n+1$$
$$s... = b_1 \leftarrow x_k + x_{n+1}$$
$$s... = b_1 \leftarrow b_1 == N$$
$$s... = y_o \leftarrow y_o \vee b_1$$

$$s^+ = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\}$$
$$\hat{s}^+ = \{y_o \leftarrow y_o \vee (x_k + x_{n+1} == N) \ \ \forall k < n+1 | \, b_1, y_o\}$$
$$s^+_{n+1} = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\} \ \cup$$
$$\{y_o \leftarrow y_o \vee (x_k + x_{n+1} == N) \ \ \forall k < n+1 | \, b_1, y_o\}$$

$$s^+_{n+1} = s^+ \cup \hat{s}^+ = P[X_{n+1}] \supseteq P[X_n] = s^+$$

26.4 Determine $O[n], O_S[n], O_T[n], f_{n+1}[n], f^T_{n+1}[n], f^S_{n+1}[n]$ for the above solution

$$O_S[n] = |y_o| + |b_1| = 2$$
$$O_T[n] = 3n(n-1) + 1 = 3n(n-1) - 1 + O_S[n]$$
$$O[n] = 3n(n-1) + 3 = 3n^2 - 3n + 3$$

$$f^S_{n+1}[n] = 0$$
$$f^T_{n+1}[n] = 6n$$
$$f_{n+1}[n] = f^S_{n+1}[n] + f^T_{n+1}[n]$$

26.5 Verify $O[n+1] = O[n] + f_{n+1}[n]$

$$O[n+1] = O[n] + \hat{O}[n]$$
$$3(n+1)^2 - 3(n+1) + 3 = 3n^2 - 3n + 3 + 6n$$
$$3n^2 + 6n + 3 - 3n - 3 + 3 = 3n^2 + 3n + 3$$

$$3n^2 + 3n + 3 = 3n^2 + 3n + 3$$

26.6 Show s^+ has Polynomial Complexity by the definition of Total Polynomial Complexity

$$O[n] = 3n^2 - 3n + 3$$

26.7 Show the limit $_{n\to\infty}\frac{O[n+1]}{O[n]}$ does not Diverge

$$limit_{n\to\infty}\frac{O[n+1]}{O[n]} =$$

$$limit_{n\to\infty}\frac{3n^2+3n+3}{3n^2-3n+3} =$$

$$limit_{n\to\infty}(\frac{3n^2-3n+3}{3n^2-3n+3} + \frac{6n}{3n^2-3n+3}) =$$

$$limit_{n\to\infty}(1 + \frac{6n}{3n^2-3n+3}) = 1$$

27 The Knapsack Problem

27.1 The Knapsack Problem

The Knapsack Problem is a famous problem in computer science which asks if objects can be stored in a knapsack. Typically the problem is designed with two constraints, weight and value. Given objects x_i, each with a respective weight w_i and value v_i, does there exist a combination of objects lighter than input weight W and more valuable than input value V?

27.2 Formal Definition

$$X_n = \{x_1, x_2, ..., x_n\} = \{\{w_1, v_1\}, \{w_2, v_2\}, ..., \{w_n, v_n\}\}$$

$$I = \{i_1, i_2, ..., i_n\} : i_l \in \{0, 1\} \ \ \forall i_l \in I$$

$$D := f[X_n, W, V] = a_o \in \{\mathbb{T}, \mathbb{F}\} = \exists I :$$
$$(\textstyle\sum_{j=1}^{n} i_j w_j < W) \wedge (\sum_{j=1}^{n} i_j v_j \geqslant V)$$

27.3 Solve for C[n]

27.3.1 Expressing I as a binary number

$$I = \{i_1, i_2, ..., i_n\} : i_l \in \{0, 1\} \ \ \forall i_l \in I$$

Valid combinations of I

$$I_{valid} =$$
$$\{\{0, 0, 0, ..., 0, 0, 1\}, \{0, 0, 0, ..., 0, 1, 0\}, \{0, 0, 0, ..., 0, 1, 1\}, ..., \{1, 1, 1, ..., 1, 1, 1\}\}$$

$$C[n] = |I_{valid}[n]| = 2^n - 1$$

27.3.2 Using a sum of combinations of inputs x_i

$$X_n = \{x_1, x_2, ..., x_n\} = \{\{w_1, v_1\}, \{w_2, v_2\}, ..., \{w_n, v_n\}\}$$

Valid combinations of x_i

$$X_{valid}[n] =$$

$$\{x_1\}\cup\{x_2\}\cup...\cup\{x_n\}\cup\{x_1,x_2\}\cup\{x_1,x_3\}\cup...\cup\{x_{n-1},x_n\}\cup...\cup\{x_1,x_2,...,x_n\}$$

$$= \ {}_{X_n}C_1 \cup \ {}_{X_n}C_2 \cup ... \cup \ {}_{X_n}C_n$$

$$C[n] = |X_{valid}[n]| = \sum_{j=1}^{n} \ {}_nC_j$$

27.3.3 Verify consistency

$$C[n] = |X_{valid}[n]| = |I_{valid}[n]|$$

$$= 2^n - 1 = \sum_{j=1}^{n} \ {}_nC_j = {}_nC_1 + {}_nC_2 + ... + {}_nC_n$$

$$= 2^n - 1 = 2^n - 1$$

27.4 Express a solution s^+ of the Knapsack Problem

27.5 Prove s^+ satisfies the subfunction condition of solutions

27.6 Determine $O[n], O_T[n], O_S[n], f_{n+1}[n]$

27.7 Show $s^+ \notin \mathbb{P}$

27.8 Express the Solution Space $\mathbb{S}$ for The Knapsack Problem

27.9 Prove a lower bound for all solutions $s^+ \in S^+ := O_{lower}[n]$

27.10 Prove D has Divergent Complexity

Parallel Programs

28 Workers

29 Definition of a Parallel Program

Optimal Programs

30 Optimal Complexity

31 Optimal Complexity $O_{opt}[n]$

Let Problem $\mathcal{Q}$ with solution space $\mathbb{S}^+$ and number of inputs n = $\tilde{n}$

$$O_{opt}[\tilde{n}] := O[\tilde{n}] : O[\tilde{n}] \leqslant O_i[\tilde{n}] \quad \forall s_i^+[\tilde{n}] \in \mathbb{S}^+$$

31.1 Proof of Existence

Let Problem $\mathcal{Q}$ with solution space $\mathbb{S}^+$
Prove the existence of $O_{opt}[n]$

32 Optimal Solutions

33 Polynomial Problems

34 Divergent Problems

AKA Not Polynomial Problems or Non-Polynomial Problems

Appendix

35 Domains

36 Bounded Input Bounded Output

In abstraction theory, functions have a domain (bounded set of every possible input and output)

Convergence, Divergence, Divergent Complexity

37 Convergent Functions

38 Divergent Functions

39 Polynomial Complexity - Note on λ

Property holds for $\lambda_i \geqslant 1$. Since O is integer non-decreasing it is trivial to show λ_i is an int > 0

$$\lambda_i \geqslant 1$$

References

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